WORKSHEET 5.2

Extreme Values and Critical Points

Unit 5: Analysing Functions

Key ideas

- A **critical point** is a point in the domain of f where $f'(x) = 0$ or $f'(x)$ does not exist.
- Every relative extremum occurs at a critical point, but not every critical point is an extremum.
- **Extreme Value Theorem.** A function continuous on a *closed* interval $[a, b]$ has an absolute maximum and an absolute minimum there.
- A point outside the domain of f is never a critical point.

Worked examples

Example 1 · A polynomial

Find the critical points of $f(x) = x^3 - 3x^2$.

Solution. $f'(x) = 3x^2 - 6x = 3x(x - 2)$.

$x = 0$ and $x = 2$.

Example 2 · Where the derivative fails

Find the critical point of $f(x) = x^{2/3}$ and describe f there.

Solution. $f'(x) = \frac{2}{3}x^{-1/3}$ is never zero, and does not exist at $x = 0$, which is in the domain.

$x = 0$ is a critical point; f has its minimum value 0 there.

Example 3 · Using the Extreme Value Theorem

Does $f(x) = \dfrac{1}{x - 1}$ have absolute extrema on $[2, 5]$? Find them.

Solution. f is continuous on $[2, 5]$, so the theorem applies. $f'(x) = -\dfrac{1}{(x-1)^2} < 0$: f is decreasing.

Maximum $f(2) = 1$; minimum $f(5) = \frac{1}{4}$.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

Practice

CORE

1. Find the critical point of $f(x) = x^2 - 10x$.

2. Find the critical point of $f(x) = x^4 - 4x$.

3. Find the critical points of $f(x) = \sin x$ on $(0, 2\pi)$.

4. Does the Extreme Value Theorem apply to $f(x) = \dfrac{1}{x}$ on $[-1, 1]$? Explain.

EXAM LEVEL

5. Find the critical point of $f(x) = xe^{x/2}$.

6. Find all critical points of $f(x) = x - 3x^{1/3}$.

7. Find the critical points of $f(x) = (x^2 - 4)^2$.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

8. Find the critical point of $f(x) = \dfrac{\ln x}{x}$ for $x > 0$, and the value of f there.

Question 9 · Free-response practice

Let $f(x) = x^4 - 18x^2$.

(a) Find all critical points of f.

(b) Find the value of f at each critical point.

(c) Find the absolute maximum and absolute minimum values of f on $[-1, 4]$.

AP® is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

Worked solutions

1. $f'(x) = 2x - 10 = 0$ at $x = 5$.
2. $f'(x) = 4x^3 - 4 = 0$ at $x = 1$.
3. $\cos x = 0$ at $x = \dfrac{\pi}{2}$ and $x = \dfrac{3\pi}{2}$.
4. No: f is not continuous on $[-1, 1]$ (it is undefined at 0).
5. $f'(x) = e^{x/2}\left(1 + \dfrac{x}{2}\right) = 0$ at $x = -2$.
6. $f'(x) = 1 - x^{-2/3}$: zero at $x = \pm 1$, and undefined at $x = 0$. Three critical points.
7. $f'(x) = 4x\left(x^2 - 4\right)$: $x = 0$ and $x = \pm 2$.
8. $f'(x) = \dfrac{1 - \ln x}{x^2} = 0$ at $x = e$, where $f(e) = \dfrac{1}{e}$.
9. (a) $f'(x) = 4x^3 - 36x = 4x(x - 3)(x + 3)$: $x = 0$ and $x = \pm 3$.
   (b) $f(0) = 0$ and $f(\pm 3) = 81 - 162 = -81$.
   (c) Candidates: $f(-1) = -17$, $f(0) = 0$, $f(3) = -81$, $f(4) = -32$. Maximum 0; minimum -81.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.